

ONE-MARK QUESTIONS

1. What is morphology in NLP?

Morphology in NLP is the study of how words are formed and how they can be broken into smaller meaningful units called morphemes. It deals with word structure, word formation rules, and how different forms of a word represent tense, gender, number, or meaning. Morphology helps computers understand variations of words such as “play,” “played,” and “playing.” It is an important step in processing natural language because it improves understanding, translation, and text analysis.

2. Define a morpheme.

A morpheme is the smallest meaningful unit of a language. It cannot be further divided into smaller parts without losing meaning. For example, words like “book,” “play,” or “happy” are single morphemes, while “unhappy” contains two morphemes: “un” (not) + “happy.” Morphemes help NLP systems understand how new words are built and how meaning changes when new parts are added.

3. What is an inflectional morpheme?

An inflectional morpheme is a type of morpheme that modifies a word to express grammatical information like tense, number, or gender without changing its core meaning or word class.

Examples include “-s” in “girls,” “-ed” in “played,” and “-ing” in “running.” These morphemes help NLP systems identify grammatical variations of words and analyze sentences correctly.

4. Define derivational morphology.

Derivational morphology is the process of creating new words by adding prefixes or suffixes that change the meaning or grammatical category of the original word. For example, “teach” becomes “teacher” (verb to noun) and “happy” becomes “unhappy” (change in meaning). Derivational morphology helps NLP systems understand relationships between root words and their derived forms.

5. What is the purpose of a morphological analyzer?

A lexicon in NLP is a digital dictionary that stores words along with their meanings, grammar rules, part-of-speech tags, and other linguistic information. It acts as a knowledge base that

helps a system understand what each word means and how it is used in a sentence. Lexicons are essential for tasks such as parsing, POS tagging, and machine translation.

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7. Define document structure.

Document structure refers to the organized arrangement of text in a document, such as headings, paragraphs, titles, lists, and sections. It helps NLP systems understand how information is grouped and how different parts relate to each other. Good document structure enables better information extraction, summarization, and classification.

Document Structure

- Title
- Headings
- Paragraphs
- Lists
- Sections

NLP

Natural language
processing

8. What is syntactic parsing?

Syntactic parsing is the process of analyzing the grammatical structure of a sentence. It identifies how words relate to each other and how they form phrases and clauses. The output is usually a parse tree that shows sentence structure. Parsing helps NLP systems understand sentence meaning, detect relationships, and perform tasks like translation, question answering, and grammar checking.

9. Define a Treebank.

A Treebank is a linguistic database that contains sentences annotated with their syntactic or semantic structure. The most common form is a syntactic Treebank where sentences are represented using parse trees. Treebanks are used to train and evaluate parsing algorithms and improve the accuracy of language models.

10. What is constituency parsing?

Constituency parsing is a type of syntactic parsing that breaks a sentence into hierarchical groups called constituents, such as noun phrases and verb phrases. It represents the structure in the form of a constituency parse tree. This helps NLP systems understand how words group together to form meaningful units in a sentence.

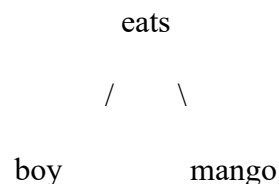
Example sentence: "The cat sleeps."



11. What is dependency parsing?

Dependency parsing is a method of analyzing a sentence by identifying how each word depends on or relates to another word. Instead of grouping words into phrases, it forms links between words called dependencies. Each dependency connects a *head* word with its *dependent*. This helps NLP systems understand grammatical relations like subject, object, and modifiers, which improves translation, question answering, and semantic analysis.

Sentence: "The boy eats mango."



The

12. Define syntactic ambiguity.

Syntactic ambiguity occurs when a sentence can be interpreted in more than one grammatical way. This happens because the structure of the sentence allows multiple possible parse trees. For example, “I saw the man with a telescope” can mean either *I used a telescope* or *the man had a telescope*. This ambiguity makes sentence interpretation difficult for NLP systems and requires advanced parsing to resolve.

13. What is a parsing algorithm?

A parsing algorithm is a computational method used to analyze the grammatical structure of a sentence. It takes a sentence as input and produces a parse tree or dependency structure. Parsing algorithms follow grammar rules like Context-Free Grammar to determine valid structures. Examples include CYK, Earley, and shift-reduce parsers. These algorithms help NLP applications understand sentence meaning, detect errors, and perform translation.

14. Define semantic parsing.

Semantic parsing is the process of converting natural language into a machine-understandable meaning representation. Instead of just analyzing grammar, it focuses on extracting the actual meaning behind a sentence. Semantic parsing links words to entities, relationships, actions, and concepts. It is widely used in chatbots, virtual assistants, question answering, and knowledge extraction.

15. What is semantic interpretation?

Semantic interpretation is the process of assigning meaning to words, phrases, and sentences in a given context. It ensures that the intended message is understood correctly. For example, understanding whether “bank” means a financial institution or a river bank depends on context. Semantic interpretation helps NLP systems perform tasks like text understanding, summarization, and machine translation.

16. Define word sense.

Word sense refers to the specific meaning of a word in a particular context. Many words have multiple meanings; for example, “bat” can mean an animal or a sports instrument. Identifying

the correct sense is important for accurate interpretation. Word sense is central to NLP tasks like translation, question answering, and semantic analysis.

17. What is WSD (Word Sense Disambiguation)?

Word Sense Disambiguation (WSD) is the process of automatically identifying which meaning of a word is being used in a particular sentence. It uses context, surrounding words, dictionaries, and machine learning models to determine the correct sense. WSD improves translation accuracy, search relevance, information extraction, and chatbots' understanding.

18. What is a predicate in NLP?

A predicate in NLP represents the action or state described by a verb in a sentence. It tells what is happening and often connects to the participants involved, such as the subject or object. For example, in "The cat sleeps," the predicate is "sleeps." Predicates are important for semantic role labeling, meaning representation, and event extraction.

19. Define argument structure.

Argument structure describes how a predicate (usually a verb) connects to its arguments, such as subject, object, and other required components. For example, the verb "give" has three arguments: giver, receiver, and object. Understanding argument structure helps NLP systems identify roles in a sentence and extract meaningful relationships between entities.

For verb "give": give → (Agent, Receiver, Theme)

20. What is a meaning representation?

Meaning representation is a structured way of expressing the meaning of a sentence so that a computer can understand it. It converts natural language into logical forms, semantic graphs, or frames that capture entities, actions, relationships, and intentions. This representation helps in tasks like reasoning, answering questions, and interacting with knowledge bases.

Sentence → Meaning Representation

"John eats apple"

→ Eat(John, Apple)

21. Define discourse.

Discourse refers to any connected sequence of sentences or spoken language that conveys a complete idea or message. It goes beyond individual sentences and focuses on how sentences relate to each other in a paragraph, conversation, or document. Discourse helps NLP systems understand context, continuity, and the flow of information. Examples include dialogues, stories, reports, and articles.

22. What is cohesion in discourse?

Cohesion refers to how sentences in a text are linked together through words, references, connectors, and repeated ideas. It creates smooth flow and clarity in communication. Words like “he,” “this,” “therefore,” or repeated keywords help maintain cohesion. In NLP, cohesion is important for summarization, text classification, and discourse analysis.

Sentence 1 → refers to → Sentence 2 → continues → Sentence 3

23. What is reference resolution?

Reference resolution is the process of identifying which real-world entity a word or phrase refers to in a text. For example, in “John lost his keys. He is worried,” the word “he” refers to John. NLP systems use reference resolution to correctly understand pronouns, repeated names, and descriptions. It improves tasks like question answering, summarization, and dialogue understanding.

24. Define an n-gram.

An n-gram is a sequence of n words (or tokens) taken together from a given text. For example, “I am” is a bigram (2 words) and “I am happy” is a trigram (3 words). N-grams help NLP models understand local patterns, predict the next word, and build language models. They are widely used in text prediction, speech recognition, and statistical analysis.

25. What is a language model?

A language model is a system that predicts the next word in a sentence based on the previous words. It learns patterns, grammar, and structure from large amounts of text. Language models estimate the probability of word sequences and are used in applications like autocorrect, speech recognition, translation, and chatbots.

26. What is perplexity?

Perplexity is a measure of how well a language model predicts the next word in a sentence. A lower perplexity score means the model is better at prediction. It evaluates how “confused” or “uncertain” a model is when processing text. Perplexity is commonly used to compare the performance of different language models.

27. What is parameter estimation in NLP?

Parameter estimation refers to the process of calculating the probabilities or weights used by NLP models. For example, in n-gram models, it estimates how often each word sequence occurs. In machine learning models, parameter estimation finds the best values for weights that reduce errors. It helps the model learn linguistic patterns from data.

28. What is the role of smoothing in n-gram models?

Smoothing is a technique used in n-gram models to handle the problem of zero probabilities for unseen word combinations. It adjusts the probability distribution so that rare or unseen sequences get a small, non-zero probability. This improves model accuracy and prevents failure during prediction. Common methods include Laplace, Good-Turing, and backoff smoothing.

29. Define backoff in language modeling.

Backoff is a method used in n-gram language models where, if a higher-order n-gram probability is unavailable, the model “backs off” to a lower-order n-gram. For example, if a trigram is unseen, the model uses the bigram probability instead. This ensures reliable word prediction even when data is limited.

Trigram missing → use Bigram

Bigram missing → use Unigram

30. What is multilingual NLP?

Multilingual NLP refers to the development of language processing systems that can understand, translate, or generate text in multiple languages. It uses multilingual datasets, cross-lingual embeddings, and models like mBERT or XLM-R. This field enables applications like multilingual chatbots, translation systems, and cross-language search engines.

31. Define cross-lingual language modeling.

Cross-lingual language modeling focuses on building models that can understand and process multiple languages by sharing knowledge between them. It allows a model trained in one language to perform tasks in another language without requiring large training data for every language. Models like XLM and mBERT learn shared multilingual representations, enabling translation, multilingual search, and cross-language text understanding.

32. What is tokenization?

Tokenization is the process of breaking text into smaller units called tokens. These tokens can be words, subwords, sentences, or characters. For example, “I love NLP” becomes ["I", "love", "NLP"]. Tokenization is the first step in NLP because models can process text only after it is split into meaningful units. Modern tokenizers also handle punctuation, special characters, and multilingual text.

33. Define morphological segmentation.

Morphological segmentation is the process of dividing a word into its meaningful morphemes, such as prefixes, root words, and suffixes. For example, “unhappiness” can be segmented as: un + happy + ness. This helps NLP systems understand word formation, derive meanings, and improve tasks like translation, speech processing, and information retrieval.

34. What is a context-free grammar?

A Context-Free Grammar (CFG) is a set of rules used to describe the structure of sentences in a language. Each rule shows how symbols can be expanded to form valid sentences. For example: $S \rightarrow NP VP$. CFGs are used in syntactic parsing, programming language compilers, and generating parse trees. They help NLP systems understand sentence structure using formal grammar rules.

CFG Example:

$S \rightarrow NP VP$

$NP \rightarrow Det N$

$VP \rightarrow V NP$

35. What is a corpus?

A corpus is a large collection of text or speech data used for training and evaluating NLP models. It contains real-world language examples such as books, articles, conversations, or web data. Corpora help NLP systems learn grammar patterns, vocabulary usage, and semantic relationships. Examples include the Brown Corpus, Penn Treebank, and Wikipedia datasets.

36. Define semantic role.

A semantic role represents the function played by an entity in a sentence with respect to the main action or verb. For example, in “John gave Mary a book,” John is the *Agent*, Mary is the *Receiver*, and the book is the *Theme*. Semantic roles help NLP models understand who did what to whom, improving text understanding and information extraction.

37. What is discourse coherence?

Discourse coherence refers to the logical and meaningful connection between sentences in a text. A coherent discourse flows smoothly and maintains a clear theme or topic. Coherence is achieved through consistent references, transitions, and related ideas. NLP uses discourse coherence to perform summarization, topic modeling, and story understanding.

38. What is the purpose of Treebanks?

Treebanks store sentences annotated with syntactic or semantic structures, providing supervised data for training NLP models. They help researchers develop and evaluate parsing algorithms, build grammar rules, and improve language models. Treebanks also serve as benchmarks for comparing parsing accuracy across different systems.

39. Define n-gram probability.

N-gram probability measures how likely a sequence of n words appears in a language. It is calculated based on frequency counts from a corpus. For example, the probability of a bigram like “good morning” depends on how often it appears compared to all bigrams. N-gram probability helps language models predict the next word and analyze text patterns.

40. What is a unigram model?

A unigram model is the simplest type of language model that treats each word as independent of the others. It calculates the probability of each word based only on how frequently it appears in a corpus. Although it ignores context, it is useful for simple tasks such as text classification and basic probability estimation. It also forms the base for higher-order n-gram models.

41. What is a trigram model?

A trigram model is a type of language model that predicts a word based on the previous two words. It uses 3-word sequences (trigrams) to estimate probabilities. For example, the probability of “going” in “I am going” is calculated using the bigram “I am.” Trigram models capture more context than unigram and bigram models, making predictions more accurate in tasks like speech recognition and text generation.

42. What is language model adaptation?

Language model adaptation refers to modifying or retraining a language model so that it performs better on a specific domain, topic, or application. For example, adapting a general English model for medical or legal text. Adaptation improves accuracy because the model learns vocabulary, style, and patterns that are specific to the target domain. Techniques include fine-tuning, vocabulary expansion, and data augmentation.

43. Define smoothing.

Smoothing is a statistical technique used to handle zero-frequency problems in language models. If a word sequence never appears in the training data, its probability becomes zero, which is harmful for prediction. Smoothing assigns a small non-zero probability to such unseen phrases. This helps create a more reliable and robust language model.

44. What is L1 smoothing?

L1 smoothing, also known as Laplace smoothing, is a method in which 1 is added to the count of every word or n-gram before calculating probabilities. This ensures that no word sequence has zero probability. Although simple, it helps make the model more stable, especially when data is limited. It is commonly used in naive Bayes and basic n-gram models.

45. What is bigram probability?

Bigram probability is the likelihood of a word occurring immediately after another word. It is calculated using 2-word sequences (bigrams). For example, the probability of “morning” after “good” depends on how often “good morning” appears in the corpus. Bigram models help capture simple context patterns and are used in prediction and text generation.

46. What is language model evaluation?

Language model evaluation measures how well a model predicts word sequences or understands language patterns. The most common metric is **perplexity**, which shows how confused the model is. Lower perplexity means better performance. Evaluation helps compare different models and ensures they work accurately for tasks like translation, speech recognition, and text prediction.

47. What does LM stand for in NLP?

In NLP, LM stands for **Language Model**. It refers to a computational model that learns patterns, grammar, and structure from text to predict or generate sentences. LMs are used in chatbots, speech recognition, translation, and text completion systems.

48. What is POS tagging?

POS (Part-of-Speech) tagging is the process of assigning grammatical categories such as noun, verb, adjective, adverb, etc., to each word in a sentence. For example, in "She runs fast," "She" is a pronoun, "runs" is a verb, and "fast" is an adverb. POS tagging helps NLP models understand sentence structure and improves parsing, information extraction, and question answering.

49. What is ambiguity in natural language?

Ambiguity occurs when a word, phrase, or sentence has more than one possible meaning. For example, "bank" may mean a river bank or a financial bank. Natural languages are full of ambiguity, making interpretation difficult for machines. NLP systems use context, grammar, and semantic information to resolve ambiguity and understand the intended meaning.

50. Define parsing efficiency.

Parsing efficiency refers to how quickly and accurately a parsing algorithm can analyze the grammatical structure of sentences. A highly efficient parser reduces computation time, handles large datasets, and produces correct parse trees. Improving parsing efficiency is important for real-time NLP applications like voice assistants, translation systems, and chatbots.

50 THREE-MARK QUESTIONS

1. Explain the types of morphemes with examples.

A morpheme is the smallest unit of meaning in a word. There are free morphemes that can stand on their own like “book” or “run”. There are bound morphemes that cannot stand alone and must attach to other words such as “-ing”, “-s”, or “un-”. Morphemes also include derivational ones that form new words like “happy” to “happiness”, and inflectional ones that change grammatical form like “walk” to “walked”.

2. Write short notes on derivational and inflectional morphology.

Derivational morphology creates new words by adding prefixes or suffixes that change meaning or part of speech, such as “teach” becoming “teacher”. Inflectional morphology does not create new words but only changes grammatical forms like tense or number, for example “cat” to “cats”. Derivational morphemes change meaning strongly, while inflectional morphemes keep the same basic meaning.

3. What is the importance of morphological analysis in NLP?

Morphological analysis helps computers understand how words are formed and what their basic meanings are. It reduces the number of different word forms by identifying roots and affixes. This helps many NLP tasks like search engines, machine translation, and text analysis because the system understands different forms of a word such as “play”, “played”, and “playing”.

4. Discuss the challenges in morphological processing.

Morphological processing is difficult because many languages have irregular and unpredictable word forms. Some languages like Hindi or Tamil have long and complex word structures. There is also a lot of ambiguity where the same word form can have more than one meaning. Because of these reasons, computers find it hard to split words correctly into morphemes.

5. Explain the concept of word formation rules.

Word formation rules explain how new words are made by joining morphemes in a correct pattern. These rules show how derivational forms like “happy” becoming “happiness” or inflectional forms like “walk” becoming “walked” are created. In NLP, these rules help computers guess the structure of new or unknown words.

6. What are the major components of a document structure?

A document is usually made of a title, headings, paragraphs, sections, and a conclusion. It may also contain metadata, references, and formatting like fonts or spacing. These components help both humans and machines understand the flow, purpose, and organization of the content.

7. Compare rule-based and statistical morphology.

Rule-based morphology depends on manually created linguistic rules to break words or add affixes. Statistical morphology uses large text data to learn patterns and probabilities. Rule-based systems are more controlled but rigid, while statistical systems handle new or irregular words better because they learn from data.

8. Explain the role of Treebanks in syntax analysis.

A Treebank is a collection of sentences that are already labeled with their grammatical structure. These annotated trees help train computer parsers to understand grammar automatically. Treebanks make it easier to compare, test, and improve different syntactic analysis systems.

9. Compare constituency and dependency parsing.

Constituency parsing breaks a sentence into phrases like noun phrases and verb phrases. Dependency parsing focuses on relations between words, where each word depends on a main head word. Constituency shows phrase structure, while dependency shows word-to-word grammatical connections.

10. Discuss common parsing difficulties in natural languages.

Parsing is difficult because natural language has many ambiguities where the same sentence can have more than one meaning. Some sentences have long-distance relationships that are hard to connect. Free word order languages and unknown words also make parsing harder for computers.

11. Explain ambiguity resolution in parsing.

Ambiguity is resolved using grammar rules, statistical models, and context information. The system checks which structure is more likely based on previous examples. Semantics and sentence meaning also help choose the correct interpretation when many possibilities exist.

12. What is the role of syntactic structure in NLP?

Syntactic structure helps show how words connect in a sentence and what roles they play. It is important for tasks like translation, question answering, and summarization because without grammar structure, the meaning of the sentence may be misunderstood. Proper syntactic analysis reduces confusion and improves accuracy.

13. Describe the steps involved in semantic parsing.

Semantic parsing first performs tokenization and syntactic parsing to understand the sentence structure. It then assigns semantic roles and relationships to identify who is doing what. Finally, it converts the sentence meaning into a logical or structured form that computers can use for reasoning.

14. Explain different system paradigms for semantic parsing.

Semantic parsing can use rule-based symbolic systems that depend on predefined grammar. It can also use statistical or machine learning systems that learn patterns from large datasets. Modern systems often combine both approaches to improve flexibility and accuracy.

15. Describe word sense systems with examples.

Word sense systems help identify which meaning of a word is used in a particular sentence. For example, the word “bank” can mean a riverbank or a financial bank. Systems like WordNet organize words into different meanings, synonyms, and relations to help computers understand context.

16. What are semantic roles? Explain with examples.

Semantic roles describe how participants in a sentence are involved in an action. For example, in “Ram cut the fruit with a knife”, Ram is the agent doing the action, the fruit is the object affected, and the knife is the instrument used. These roles help computers understand sentence meaning clearly.

17. What is meaning representation? Give types.

Meaning representation is a way to express the meaning of sentences in a form that computers can understand. Types include logic-based representations, frame-based structures, semantic networks, and conceptual graphs. These help computers perform reasoning, answering, and decision-making

18. Explain the need for discourse processing.

Discourse processing is needed because meaning does not come only from single sentences; it comes from how sentences connect in a passage. It helps understand context, topic flow, and references like pronouns. It is useful in applications such as chatbots, conversation systems, and summarizers.

19. What is cohesion? Explain types with examples.

Cohesion refers to the grammatical links that join sentences together. It includes reference words like “he” or “this”, conjunctions like “and” or “therefore”, and repetition or synonyms of words. For example, “Sita came home. She was tired” uses cohesive reference.

20. Explain reference resolution challenges.

Reference resolution is difficult because pronouns like “he”, “they”, or “it” may refer to different possible nouns. Sometimes the actual referent is far apart in the text. Understanding real-world knowledge is also required, which makes automatic reference resolution challenging.

21. Describe the n-gram model with examples.

An n-gram model predicts a word based on the previous words. A unigram ignores context, a bigram looks at one previous word, and a trigram looks at two previous words.

For example, in the phrase “go to school”, a bigram model predicts “school” after “to”.

22. What is perplexity? Explain its significance.

Perplexity is a measure of how well a language model predicts text. Lower perplexity means the model is more accurate and confident. It is used to compare different language models and select the best performing one.

23. Explain the importance of language model adaptation.

Language model adaptation helps a model adjust to a specific domain such as medical, legal, or technical text. When a model is trained on general data, it may not perform well in specialized areas. Adaptation improves accuracy by learning domain-specific terms and patterns.

24. Discuss parameter estimation in n-gram models.

Parameter estimation involves calculating probabilities using frequency counts from a corpus. For example, $P(\text{word} \mid \text{previous word})$ is computed by dividing the count of the bigram by the count of the previous word. Smoothing techniques are used when data is missing.

25. Compare monolingual and multilingual language models.

Monolingual language models are trained in one language and achieve high accuracy for that language. Multilingual language models are trained on many languages together and allow cross-lingual understanding. However, multilingual models may be slightly less accurate for individual languages.

26. What are language-specific modeling problems?

Some languages have very rich morphology or flexible word order, making prediction difficult. Scripts with complex writing systems also create challenges. These differences require special handling because one model cannot fit all languages equally well.

27. Discuss multilingual issues in parsing.

Multilingual parsing is difficult because each language has its own grammar rules and sentence patterns. Code-switching, where people mix languages, adds more confusion. Many languages lack sufficient annotated data, making training hard for parsers.

28. Explain structured vs. unstructured discourse.

Structured discourse is organized clearly with sections and logical flow, such as essays or reports. Unstructured discourse, like casual conversations or social media posts, is informal and unpredictable. Structured text is easier for NLP systems to process compared to unstructured text.

29. What is Treebank annotation? Explain types.

Treebank annotation adds syntactic details to sentences in the form of trees. It can include constituency trees that show phrases, or dependency trees that show head-dependent relations. Some Treebanks also include semantic or morphological information.

30. Explain neural language models in brief.

Neural language models use neural networks to learn relationships between words. Models like RNNs, LSTMs, and Transformers capture long-range dependencies better than old statistical models. They are widely used in modern NLP systems like BERT and GPT.

31. What is backoff? Give examples.

Backoff is used when higher n-gram counts are zero. If a trigram is missing, the system shifts to a bigram or unigram probability. This helps avoid zero probabilities and improves prediction accuracy.

32. Describe the role of smoothing in improving language models.

Smoothing helps when some n-grams do not appear in the training data. It distributes small probability values to unseen words. This makes the model more reliable and avoids zero-probability errors.

33. Discuss different types of language models.

Language models include statistical models like unigram, bigram, and trigram, and neural models like RNN, LSTM, and Transformer-based systems. Some models also combine both approaches or work across multiple languages.

34. Explain cohesion vs. coherence.

Cohesion refers to the grammatical connections between sentences, while coherence refers to the logical flow of ideas. A text may be cohesive but still lack coherence if ideas are not logically arranged.

35. What is discourse structure? Explain with examples.

Discourse structure refers to how a text is organized into meaningful parts such as introduction, body, and conclusion. Rhetorical relations like cause, contrast, or explanation connect these parts. For example, an essay discussing a problem followed by its solution shows discourse structure.

36. Compare top-down and bottom-up parsing.

Top-down parsing starts from the highest-level grammar rule and breaks it into smaller parts. Bottom-up parsing starts from the words in the sentence and builds up to the complete structure. Top-down focuses on expectations, while bottom-up focuses on data.

37. What is dependency grammar?

Dependency grammar represents sentences through relations where words depend on a main head word. These dependencies form a tree showing how words connect. It is very useful for languages that allow free word order.

38. Describe challenges in morphological segmentation.

Segmentation is hard because many languages have irregular forms and complex affixes. Some words can be split in more than one correct way, causing ambiguity. Agglutinative languages create extremely long words that are difficult to break.

39. Explain the role of software tools in semantic analysis.

Tools like WordNet, spaCy, and semantic parsers help computers understand word meanings, relations, and roles. They make tasks like translation, question answering, and text classification more accurate. These tools support the process of converting text into meaningful structures.

40. Discuss the challenges of cross-lingual language modeling.

Cross-lingual modeling is hard because languages differ in grammar, script, and vocabulary. Cultural meanings and idioms do not translate easily. Many languages also lack enough parallel data for training.

41. Write short note on word sense disambiguation.

Word sense disambiguation selects the correct meaning of a word according to its context. For example, “bat” can be an animal or sports equipment. The system uses context clues and knowledge resources to decide the right meaning.

42. Explain n-gram probability computation.

N-gram probability is calculated using word frequency counts from a corpus. For example, the trigram probability of three words is the count of all three words appearing together divided by the count of the first two. This helps the model choose the most likely next word.

43. What is discourse anaphora?

Discourse anaphora refers to using expressions like pronouns that point back to something mentioned earlier. For example, “Meera bought a dress. She loved it.” Here “she” refers to Meera. Resolving anaphora is important for understanding text flow.

44. Explain reference chains with examples.

Reference chains connect all mentions of the same person or thing across a text. For example, “Rahul came home. He sat down. The boy was tired” all refer to Rahul. These chains help maintain continuity and meaning.

45. Discuss the concept of semantic roles.

Semantic roles identify the function of each noun in a sentence, such as the doer, receiver, or instrument of an action. They help computers understand the meaning by showing who is doing what. This is important for tasks like semantic parsing and machine translation.

46. Compare symbolic and statistical semantic parsing.

Symbolic parsing uses hand-written rules and grammar to analyze sentence meaning, but it may fail on new or complex sentences. Statistical parsing learns patterns from data and is more flexible. Modern systems prefer a combination of both approaches.

47. Explain unigram vs. bigram vs. trigram models.

In a unigram model each word is predicted independently. In a bigram model the next word is predicted based on one previous word. In a trigram model two previous words are considered. Higher n gives more context but needs more data.

48. What are the advantages of Treebanks?

Treebanks provide a large collection of sentences with correct grammatical labels. They help train and test parsers to understand grammar better. Treebanks also allow researchers to evaluate and compare NLP systems.

49. Explain the importance of discourse markers.

Discourse markers like “however”, “therefore”, and “meanwhile” help show relationships between ideas. They make text smoother and easier to understand. NLP systems use these markers to analyze the structure and meaning of a passage.

50. What are the core challenges in NLP parsing?

Parsing is challenging because natural language is full of ambiguity and complex sentence patterns. Different languages have different grammar structures, which makes universal parsing difficult. Unknown words and long-distance dependencies also create problems for parsers.

50 FIVE-MARK QUESTIONS

1. Explain morphological models in NLP with detailed examples.

Morphological models in NLP are systems designed to analyze and understand the structure of words. They help identify roots, prefixes, suffixes, and inflectional or derivational forms of words. Morphology is important because a single word can have multiple forms, such as "run," "runs," "running," and "ran," which all share the root "run." Morphological models are generally divided into three types: rule-based, statistical, and neural models. **Rule-based models** rely on handcrafted linguistic rules. For example, a rule may specify that adding "-ed" forms the past tense of regular verbs. **Statistical models** use probabilities derived from large corpora to predict likely word forms. For example, if “cats” appears frequently with the base form “cat,” the model can learn this relation. **Neural models** use machine learning techniques like recurrent neural networks or transformers to capture complex morphological patterns and long-distance dependencies in languages. For example, a neural morphological analyzer can predict that “unbelievable” consists of “un- + believe + -able” even if it has never seen the exact word before. Morphological models are applied in POS tagging, lemmatization, information retrieval, and machine translation. They reduce vocabulary size, improve text normalization, and help disambiguate words. Languages like Finnish or Turkish, which are highly inflectional, benefit greatly from these models. Overall, morphological models are crucial in NLP because they allow systems to handle the rich variety of word forms in human languages.

unbelievable



| | |

un- believe -able

(prefix) (root) (suffix)

2. Discuss word structure and challenges in morphological analysis.

Word structure refers to the way morphemes combine to form words in a language. Words can have roots, prefixes, suffixes, and inflectional endings. For example, “replaying” has the prefix “re-”, the root “play,” and the suffix “-ing.” Understanding word structure is crucial for NLP systems because it allows them to identify the base meaning and grammatical function. However, morphological analysis faces several challenges. First, **irregular word forms** like “went” for “go” do not follow standard rules. Second, **highly inflectional languages** such as Turkish, Finnish, or Hindi generate many word forms from a single root, making data sparse. Third, **compound words** like “notebook” or “railwaystation” are difficult to segment correctly. Fourth, **ambiguity** arises when a morpheme could belong to more than one word, such as “s” in “books” (plural) or “she’s” (is). Finally, **derivational and inflectional morphemes** may overlap, creating additional complexity. NLP systems must carefully identify roots and affixes while handling these irregularities. Advanced approaches like statistical and neural models can handle unseen or rare words by learning patterns from large corpora. Rule-based approaches still help in low-resource languages or highly regular words. Correctly analyzing word structure improves lemmatization, POS tagging, and machine translation by reducing errors and improving understanding of sentence meaning.

3. Explain methods for identifying document structure.

Document structure defines how text is organized into sections, headings, paragraphs, and metadata. Identifying document structure is important for NLP applications like summarization, question answering, and content analysis. Several methods exist to identify document structure. **Rule-based methods** rely on patterns in formatting, such as headings in bold, numbered lists, or line breaks. For example, a rule may detect a section title if the line is in uppercase followed by a blank line. **Machine learning methods** use features like font size, indentation, punctuation, and word frequency to predict structure elements. Classifiers can be trained to recognize headings, paragraphs, and references. **Template-based approaches** rely on predefined templates, common in structured documents like scientific articles or invoices. **Graph-based methods** represent text elements as nodes and edges to detect hierarchy and relationships between sections. A major challenge is **unstructured or inconsistent documents**, such as blogs or social media posts, where headings or sections are absent. Hybrid methods combining

rules and machine learning often yield the best results. Correctly identifying document structure enables better NLP tasks, such as extracting relevant sections, creating summaries, or enabling automated information retrieval.

4. Compare rule-based, statistical, and neural morphological models.

Rule-based morphological models use handcrafted linguistic rules to analyze words. They are precise for regular words and low-resource languages but fail with irregular forms or unseen words. Statistical models use large text corpora to calculate probabilities of morpheme sequences and predict likely forms. They handle variability and unknown words better than rules but require large datasets. Neural morphological models use deep learning techniques, such as RNNs, LSTMs, or Transformers, to learn complex patterns and long-distance dependencies. They can generalize to unseen words and capture non-linear relationships but need extensive training data and computational resources. In practice, rule-based models are simple and interpretable, statistical models are robust with moderate data, and neural models excel in high-resource, complex languages. Many NLP systems combine these approaches to achieve high accuracy.

5. Explain parsing algorithms and their complexities.

Parsing algorithms analyze sentence structure to produce syntactic trees or dependency graphs. Common algorithms include **top-down parsing**, which starts from the start symbol and predicts constituents, and **bottom-up parsing**, which builds trees from input tokens. **Chart parsing** is a dynamic programming approach that stores intermediate results to avoid repeated computations. **Shift-reduce parsing** is used in dependency parsing for efficient processing. Complexity varies: naive top-down or bottom-up parsing can be exponential in sentence length, while dynamic programming reduces it to polynomial time. Probabilistic parsers add statistical models to handle ambiguity, selecting the most likely parse. Efficient parsing algorithms are critical for real-time NLP applications like speech recognition, machine translation, and question answering.

6. Write a detailed note on Treebanks and their construction.

A Treebank is a collection of sentences annotated with syntactic structure. Treebanks are essential for training and evaluating parsers. They provide detailed annotations of phrases, word categories, and dependencies. Construction involves tokenization, POS tagging, and manual or semi-automatic tree annotation. Some Treebanks include **constituency trees**, which represent phrases like NP or VP, while others use **dependency trees**, which focus on word-to-word relations. Building a Treebank requires linguistic expertise, consistency checks, and often

multiple annotators to reduce errors. Examples include the Penn Treebank for English and the Universal Dependencies Treebank for multiple languages. Treebanks support syntactic parsing, POS tagging, and semantic role labeling.

7. Discuss ambiguity resolution models in parsing.

Parsing ambiguity arises when a sentence can have multiple valid syntactic structures. Ambiguity resolution models help choose the most appropriate parse. **Rule-based models** use grammar rules and constraints to select the correct structure. **Statistical models** assign probabilities to parses using n-gram or probabilistic context-free grammar (PCFG) approaches. **Neural models** leverage deep learning to learn patterns from annotated data and resolve ambiguity based on context. Hybrid models combine rules with statistical learning for higher accuracy. Ambiguity resolution is critical in NLP applications like machine translation, question answering, and voice assistants to correctly interpret user input. Parsing ambiguity arises when a sentence can have multiple valid syntactic structures. Ambiguity resolution models help choose the most appropriate parse. **Rule-based models** use grammar rules and constraints to select the correct structure. **Statistical models** assign probabilities to parses using n-gram or probabilistic context-free grammar (PCFG) approaches. **Neural models** leverage deep learning to learn patterns from annotated data and resolve ambiguity based on context. Hybrid models combine rules with statistical learning for higher accuracy. Ambiguity resolution is critical in NLP applications like machine translation, question answering, and voice assistants to correctly interpret user input.

8. Compare top-down, bottom-up, and chart parsing.

Top-down parsing starts from the start symbol and predicts constituent expansions. It is goal-directed but inefficient with left recursion. Bottom-up parsing begins with input tokens and builds parse trees to reach the start symbol. It is data-driven and can handle left recursion better but may generate unnecessary structures. Chart parsing uses dynamic programming to store intermediate results, avoiding redundant computations. It is efficient, handles ambiguity, and can parse all context-free grammars. In practice, chart parsing is preferred for efficiency and real-time applications.

9. Explain semantic parsing architecture with diagrams.

Semantic parsing converts natural language sentences into structured meaning representations such as logical forms or database queries. The architecture typically includes **input processing**,

syntactic parsing, semantic role labeling, and meaning representation generation. Preprocessing includes tokenization and POS tagging. Syntactic parsers produce tree or dependency structures. Semantic role labeling identifies agents, patients, and instruments. The final component maps this information into logical forms or query structures. Neural architectures use encoder-decoder models to perform end-to-end semantic parsing.

Input Sentence

|

Tokenization

|

POS & Syntax Parsing

|

Semantic Role Labeling

|

Meaning Representation / Logical Form

Natural language
processing

10. Discuss system paradigms in semantic interpretation.

Semantic interpretation can be handled using **symbolic systems, statistical systems, or neural systems**. Symbolic systems rely on manually created grammars and lexicons, providing precise but rigid interpretations. Statistical systems use probabilistic models trained on annotated corpora to handle ambiguity and variation in language. Neural systems, such as sequence-to-sequence models or transformers, learn complex patterns from large datasets and can generalize to unseen sentences. Hybrid systems combine symbolic rules with statistical or neural methods to leverage the advantages of both approaches. Semantic interpretation systems are widely used in question answering, information extraction, and natural language understanding.

11. Explain word sense systems and challenges in WSD.

Word Sense Disambiguation (WSD) is the process of identifying the correct meaning of a word in context. For example, the word “bank” can mean a financial institution or the side of a river. Word sense systems use dictionaries, thesauri, or annotated corpora to determine word meanings. Approaches include **knowledge-based methods**, which rely on resources like

WordNet; **supervised machine learning methods**, which use labeled data; and **unsupervised methods**, which cluster words based on context. Challenges in WSD include **polysemy** (one word has multiple meanings), **rare words**, and **domain-specific senses**. The same word may have different meanings in news, literature, or scientific texts. Ambiguity increases with sentence complexity, idiomatic expressions, and context distance. Neural networks, especially transformer models like BERT, improve WSD by capturing contextual relationships, but high-quality labeled data is still required. Effective WSD is critical for machine translation, search engines, and question answering.

12. Describe predicate-argument structure and its importance.

Predicate-argument structure describes the relationship between verbs (predicates) and their participants (arguments). In the sentence “Alice gave Bob a book,” “gave” is the predicate, “Alice” is the agent, “Bob” is the recipient, and “a book” is the theme. Understanding these structures allows NLP systems to capture who did what to whom. This is important for tasks like semantic parsing, information extraction, and machine translation. Arguments can be identified through syntactic parsing and semantic role labeling. Complex sentences with multiple clauses or passive voice present challenges, but modern neural models can learn patterns for argument detection. Accurate predicate-argument analysis improves text understanding and enables reasoning about events in natural language.

13. Explain meaning representation systems with examples.

Meaning representation systems convert natural language into structured formats for computers. Common types include **logical forms**, **semantic networks**, **frames**, and **conceptual graphs**. For example, the sentence “John loves Mary” can be represented logically as `love(John, Mary)` or in a frame as `LOVE {Agent: John, Patient: Mary}`. These representations allow NLP systems to perform reasoning, question answering, and inference. Challenges include handling ambiguity, idioms, negation, and temporal expressions. Neural models now attempt end-to-end mapping from text to logical forms, reducing manual rule design while improving adaptability. Meaning representations are essential for building AI systems that understand and manipulate natural language.

14. Discuss cohesion, coherence, and discourse structure in detail.

Cohesion refers to the grammatical and lexical links between sentences, such as pronouns or conjunctions. For example, “Rita is a doctor. She works in a hospital” uses “she” for cohesion. Coherence is the logical flow of ideas, ensuring the text makes sense as a whole. A text may be cohesive but incoherent if ideas are out of order. Discourse structure defines the organization

of a text into sections, paragraphs, and rhetorical relations such as cause, contrast, or elaboration. Recognizing discourse structure helps NLP systems summarize text, answer questions, and maintain context in conversations. Challenges arise in informal texts, multiple topics, and ambiguous references, which require both linguistic rules and machine learning approaches for effective processing.

15. Explain reference resolution techniques and algorithms.

Reference resolution identifies which entity pronouns or noun phrases refer to in a text. Techniques include **rule-based approaches**, like Hobbs' algorithm, which traverses parse trees to find antecedents, and **statistical approaches**, which use features like gender, number, and distance. **Centering theory** models discourse coherence by predicting likely references based on focus and salience. Neural methods, especially transformers, leverage context and learn reference patterns from large corpora. Challenges include ambiguous pronouns, long-distance dependencies, and multiple entities with similar features. Accurate reference resolution is essential for machine translation, summarization, and question answering systems.

16. Describe n-gram language models, formulas, estimation, and challenges.

An n-gram language model predicts the next word based on the previous n-1 words. A **bigram model** predicts the next word using one previous word, and a **trigram** uses two previous words. Probability estimation is done using frequency counts:

$$P(\text{word}_n \mid \text{word}_{n-1}, \text{word}_{n-2}) = \text{Count}(\text{word}_{n-2}, \text{word}_{n-1}, \text{word}_n) / \text{Count}(\text{word}_{n-2}, \text{word}_{n-1})$$

Challenges include **data sparsity**, where some word sequences never appear in training data, and handling **rare or unseen words**. Smoothing techniques like Laplace or Kneser-Ney are used to assign small probabilities to unseen n-grams. N-gram models are simple and interpretable but cannot capture long-distance dependencies effectively.

17. Discuss parameter estimation for language models with worked examples.

Parameter estimation calculates the probabilities for word sequences in language models. In a bigram model, the probability of “cat sleeps” is:

$$P(\text{sleeps} \mid \text{cat}) = \text{Count}(\text{cat sleeps}) / \text{Count}(\text{cat})$$

For example, if “cat sleeps” occurs 5 times and “cat” occurs 20 times, $P(\text{sleeps} \mid \text{cat}) = 5/20 = 0.25$. When sequences are unseen, smoothing is applied. Maximum Likelihood Estimation

(MLE) is commonly used, while more advanced methods include Good-Turing and backoff models. Accurate parameter estimation ensures the language model predicts text correctly and handles sparse data efficiently

18. Compare statistical, neural, and hybrid language models.

Statistical models like n-grams rely on frequency counts from corpora and are simple but limited in capturing long-distance dependencies. Neural language models use deep learning, such as RNNs, LSTMs, or Transformers, to model complex patterns and context across sentences. They generalize better to unseen text but require large datasets and computing resources. Hybrid models combine statistical and neural approaches, leveraging interpretability of statistical models and flexibility of neural models. Choice depends on resources, language complexity, and application requirements.

19. Explain multilingual and cross-lingual language modeling challenges.

Multilingual language modeling involves training models on multiple languages, while cross-lingual modeling aims to transfer knowledge from one language to another. Challenges include **different grammar structures, scripts, polysemy across languages, and limited parallel corpora** for low-resource languages. Models must handle code-switching, idioms, and culturally specific terms. Transformers like mBERT and XLM-R address some challenges but require careful preprocessing, tokenization, and alignment for effective cross-lingual understanding.

20. Discuss the role of discourse processing in natural language understanding.

Discourse processing analyzes relationships between sentences, paragraphs, and entire texts to understand meaning beyond individual sentences. It helps in coreference resolution, summarization, sentiment analysis, and question answering. Discourse markers like “however” or “therefore” indicate relations such as contrast or cause. Cohesion and coherence are identified to maintain logical flow. Challenges include long-distance dependencies, topic shifts, and informal text structures. Effective discourse processing improves AI’s ability to understand context, maintain conversation continuity, and reason about text.

21. Explain language model evaluation methods with examples.

Language model evaluation measures how well a model predicts text. One common metric is **perplexity**, which indicates how uncertain the model is in predicting the next word. Lower perplexity means better performance. For example, if a trigram model predicts “cat sleeps” with probability 0.25, the perplexity decreases as the model becomes more confident. Another

evaluation method is **accuracy on downstream tasks** like machine translation, text generation, or question answering. Human evaluation can also be used for fluency and coherence. Challenges include evaluating models across multiple domains or languages, where training data may not fully match test scenarios. Effective evaluation ensures that language models perform reliably in real-world applications.

22. Discuss types of language models and their suitability for different tasks.

Language models can be **statistical**, **neural**, or **hybrid**. Statistical models, like unigram, bigram, and trigram models, are simple and suitable for small datasets or low-resource languages. Neural models, such as RNNs, LSTMs, and Transformers, are suitable for tasks requiring context, such as text generation, machine translation, and semantic analysis. Hybrid models combine statistical and neural techniques to leverage interpretability and contextual understanding. Choosing the right model depends on task complexity, available data, and computational resources. Statistical models are interpretable, neural models handle complex dependencies, and hybrid models balance both.

23. Explain smoothing techniques (Laplace, Good-Turing, Kneser-Ney) in detail.

Smoothing assigns small probabilities to unseen words or n-grams to prevent zero probability errors in language models. **Laplace smoothing** adds one to all counts, ensuring no zero probabilities but may overestimate rare events. **Good-Turing smoothing** adjusts probabilities of observed n-grams and redistributes the leftover probability mass to unseen n-grams based on their frequency. **Kneser-Ney smoothing** improves prediction by considering the number of contexts a word appears in, making it highly effective for bigrams and trigrams. Smoothing is essential for handling sparse data and improving model generalization in NLP tasks.

24. Compare language-specific and general-purpose language models.

Language-specific models are trained on a single language, achieving high accuracy in that language due to specialized vocabulary, grammar, and morphology handling. General-purpose models are trained on multiple languages and aim to provide cross-lingual understanding but may slightly reduce accuracy for individual languages. Language-specific models are ideal for high-resource languages with abundant data, while general-purpose models help multilingual tasks and low-resource languages. Cross-lingual transfer and domain adaptation can further improve performance in general-purpose models.

25. Explain morphological segmentation algorithms.

Morphological segmentation algorithms split words into morphemes, such as roots, prefixes, and suffixes. **Rule-based algorithms** use handcrafted rules for segmentation, effective for regular words but limited for irregular forms. **Statistical algorithms** rely on frequency and co-occurrence patterns from corpora, identifying morpheme boundaries based on likelihood. **Neural algorithms** use deep learning to predict morpheme boundaries, capturing complex and unseen forms. For example, “unbelievable” can be segmented as “un- + believe + -able.” Morphological segmentation aids lemmatization, POS tagging, and machine translation, improving NLP system accuracy.

26. Discuss approaches to parsing highly ambiguous languages.

Languages with free word order or rich inflection, like Hindi or Turkish, present parsing challenges. Approaches include **probabilistic parsers** using PCFGs to choose the most likely structure, **dependency parsing** to capture word-to-word relationships, and **neural parsers** leveraging context from large datasets. Hybrid approaches combine statistical and rule-based methods to improve accuracy. Additional strategies include morphological analysis for word forms and semantic constraints to resolve ambiguity. Handling ambiguity is crucial for machine translation and syntactic understanding.

27. Explain Treebank construction challenges in multilingual NLP.

Building Treebanks for multiple languages is difficult due to grammar differences, word order variation, and limited annotated corpora. Languages may have rich morphology or low resources, making annotation labor-intensive. Consistency across annotators is a challenge, especially with complex syntactic structures. Universal Dependencies framework helps standardize annotations, but domain-specific texts may require additional guidelines. Multilingual Treebanks are essential for training parsers and cross-lingual NLP tasks but require careful planning and validation.

28. Describe predicate logic as a meaning representation system.

Predicate logic represents sentence meaning using predicates and arguments. For example, “John loves Mary” is represented as $\text{love}(\text{John}, \text{Mary})$. It allows formal reasoning, inference, and querying. Predicate logic can handle negation, quantifiers, and relations between entities. Challenges include mapping natural language ambiguity, handling complex sentences, and temporal expressions. It is widely used in semantic parsing, question answering, and knowledge representation, providing a structured, interpretable meaning format for NLP systems.

29. Explain N-gram model adaptation techniques.

N-gram model adaptation adjusts probabilities to improve performance in new domains or datasets. Techniques include **linear interpolation**, which combines source and target domain probabilities, **backoff**, which uses lower-order n-grams when higher-order n-grams are missing, and **domain-specific retraining**, where counts are updated with domain-specific text. For example, adapting a general English trigram model to medical texts ensures terms like “blood pressure” receive higher probability. Adaptation improves accuracy and reliability in domain-specific NLP applications.

30. Discuss performance issues in document structure analysis.

Performance in document structure analysis is affected by inconsistent formatting, noisy data, and unstructured content like blogs or social media posts. Algorithms may misidentify headings, sections, or paragraph boundaries. Large datasets introduce computational challenges, requiring efficient parsing and feature extraction methods. Combining rule-based heuristics with machine learning models improves accuracy. Scalability and handling multilingual documents are also key issues. Correct analysis is critical for summarization, information retrieval, and automated document understanding.

31. Compare symbolic, statistical, and neural semantic parsing.

Symbolic semantic parsing relies on manually created rules, grammars, and lexicons to map sentences to meaning representations. It is precise and interpretable but rigid and unable to handle unseen patterns. Statistical parsing uses probabilities derived from annotated corpora to resolve ambiguity and select the most likely parse. It generalizes better than symbolic methods but requires large datasets and may produce errors in rare constructions. Neural semantic parsing uses deep learning, such as encoder-decoder models or transformers, to learn mappings directly from sentences to logical forms. Neural models handle unseen sentences, complex dependencies, and long-distance relations effectively but need substantial training data and computational power. Hybrid approaches combine symbolic rules with statistical or neural methods to leverage the strengths of all three. Choice depends on data availability, language complexity, and application requirements.

32. Explain discourse representation theory (DRT).

Discourse Representation Theory (DRT) is a framework for representing the meaning of discourse beyond single sentences. It uses **Discourse Representation Structures (DRS)** to capture entities, events, and their relationships across multiple sentences. For example, “A man

entered. He sat down” is represented in DRS with a variable for the man, linked across sentences using reference. DRT helps resolve pronouns, quantify expressions, and handle negation or temporal relations. It is widely used in semantic interpretation, anaphora resolution, and dialogue systems. The key advantage is modeling **context-dependent meaning**, allowing NLP systems to understand coherent discourse rather than isolated sentences.

33. Discuss multilingual parsing architectures.

Multilingual parsing architectures aim to process multiple languages using shared or language-specific components. Approaches include **universal parsing frameworks**, such as Universal Dependencies (UD), which standardize syntactic annotations across languages, and **multilingual neural parsers**, trained on parallel corpora to handle several languages simultaneously. Challenges include varying word order, rich morphology, and low-resource languages with limited annotated data. Techniques like transfer learning, cross-lingual embeddings, and multi-task learning improve parsing accuracy. Multilingual parsers are essential for machine translation, cross-lingual information retrieval, and multilingual NLP applications.

34. Explain reference resolution algorithms (Hobbs, Centering theory).

Reference resolution algorithms identify which entities pronouns or noun phrases refer to. **Hobbs’ algorithm** traverses parse trees in a left-to-right manner, selecting the first compatible antecedent based on syntactic constraints. **Centering theory** models discourse coherence by ranking entities based on salience and focus, predicting likely referents. Neural methods extend these algorithms by using context embeddings and attention mechanisms to resolve complex or long-distance references. Effective reference resolution is vital for coreference resolution, machine translation, summarization, and dialogue systems.

35. Discuss cross-lingual language model training methods.

Cross-lingual language models are trained to understand and process multiple languages simultaneously. Methods include **joint training** on parallel corpora, **transfer learning**, where models trained on high-resource languages are fine-tuned on low-resource ones, and **multilingual embeddings**, aligning words with similar meaning across languages. Challenges include varying scripts, grammar, and limited parallel data. Models like mBERT, XLM-R, and multilingual GPT leverage these techniques for tasks such as cross-lingual question answering and machine translation.

36. Explain coherence relations and their role in discourse.

Coherence relations describe logical connections between sentences or clauses, such as **cause-effect, contrast, elaboration, and temporal order**. For example, “It rained heavily. Therefore, the match was canceled” uses a cause-effect relation. Identifying coherence relations allows NLP systems to understand text flow, generate summaries, and answer questions accurately. Challenges include implicit relations, ambiguous connectors, and multi-clause sentences. Machine learning and neural models often learn coherence patterns from annotated corpora to improve discourse understanding.

37. Compare CFGs, dependency grammars, and neural parsers.

Context-Free Grammars (CFGs) define rules for phrase structure, representing hierarchical sentence structure. They are interpretable but may not capture long-distance dependencies effectively. **Dependency grammars** focus on relationships between words, capturing who does what to whom; they handle free word order languages better. **Neural parsers** learn syntactic and semantic patterns directly from data using deep learning, generalizing to unseen sentences and complex constructions. CFGs are suitable for rule-based parsing, dependency grammars for structured syntactic analysis, and neural parsers for high-performance, data-driven NLP applications.

38. Explain the integration of semantic and syntactic parsing.

Integration of semantic and syntactic parsing combines grammatical analysis with meaning representation. Syntactic parsing produces tree structures or dependencies, which serve as input for semantic parsing. Semantic role labeling then identifies predicates, agents, patients, and other roles. Integrated models improve NLP tasks like question answering, information extraction, and machine translation. Neural architectures often perform end-to-end parsing, capturing both syntax and semantics simultaneously, reducing error propagation. This integration ensures accurate understanding of both structure and meaning in natural language.

39. Discuss the importance of discourse markers with analysis.

Discourse markers are words or phrases like “however,” “therefore,” “meanwhile,” and “because” that indicate relations between sentences. They signal contrast, causality, temporality, or elaboration, guiding readers through coherent text. Identifying discourse markers helps NLP systems detect coherence, summarize content, and generate natural responses in dialogue systems. Challenges include implicit markers, ambiguous usage, and context-dependent interpretation. Combining rule-based detection with machine learning improves marker recognition and enhances overall text understanding.

40. Explain the evolution of language models from n-gram to transformers.

Language models evolved from **n-gram models**, which predict words based on limited context, to **neural models** like RNNs and LSTMs that capture long-distance dependencies, and finally to **transformers** like BERT and GPT. N-grams are simple but struggle with sparse data and long-range dependencies. RNNs and LSTMs handle sequences better but suffer from vanishing gradients. Transformers use self-attention to process all words in context simultaneously, enabling parallel training, better contextual understanding, and superior performance in NLP tasks like translation, summarization, and question answering. This evolution reflects increasing model complexity, data requirements, and real-world effectiveness.

41. Compare LM adaptation for domain-specific applications.

Language Model (LM) adaptation adjusts a general-purpose language model to perform better in a specific domain, like medicine, finance, or law. Techniques include **fine-tuning**, where the model is trained on domain-specific text; **linear interpolation**, combining general and domain-specific probabilities; and **backoff methods**, which use lower-order n-grams when domain-specific sequences are sparse. Domain-specific adaptation improves vocabulary coverage, phrase prediction, and contextual understanding. For example, a general English model may not predict “myocardial infarction” accurately, but adaptation with medical text enhances performance. Challenges include balancing general and domain-specific knowledge, avoiding overfitting, and ensuring sufficient domain data. Proper LM adaptation is crucial for specialized NLP applications such as chatbots, question-answering systems, and technical document processing.

42. Discuss challenges in building discourse-aware language models.

Discourse-aware language models analyze context across sentences, paragraphs, or entire documents. Challenges include **long-distance dependencies**, where the model must remember entities or events mentioned earlier; **topic shifts**, which can confuse coherence tracking; and **pronoun or reference resolution**, especially in complex texts. Annotated corpora for discourse are limited, making supervised training difficult. Handling multiple discourse relations, such as causality, contrast, or elaboration, requires sophisticated architectures like transformers with attention mechanisms. Informal or noisy texts, like social media posts, add additional complexity. Overcoming these challenges ensures AI systems maintain context, produce coherent text, and understand narrative structure effectively.

43. Explain multilingual sense disambiguation systems.

Multilingual sense disambiguation systems identify the correct meaning of words across languages. They extend traditional WSD methods to handle polysemy, context variation, and translation ambiguity. Approaches include **knowledge-based methods**, using multilingual lexicons like WordNet; **supervised methods**, trained on parallel corpora; and **neural methods**, leveraging multilingual embeddings or transformers (e.g., mBERT, XLM-R). For example, the English word “bank” may translate to French as “banque” (financial) or “rive” (river), requiring context for accurate mapping. Challenges include low-resource languages, inconsistent sense inventories, and cultural differences. Accurate multilingual WSD is essential for translation, cross-lingual search, and global NLP systems.

44. Compare bottom-up vs top-down approaches in semantic parsing.

Bottom-up semantic parsing builds meaning representations starting from words or phrases and combining them to form higher-level structures. It is data-driven and flexible but may produce invalid global interpretations if local decisions are incorrect. Top-down parsing starts from the highest-level structure and decomposes it into substructures, ensuring global coherence but requiring accurate grammar or rules. Hybrid approaches combine both, using top-down constraints and bottom-up evidence. Neural models often integrate these strategies using attention mechanisms, balancing local context and global structure. Choosing an approach depends on the complexity of language, available resources, and desired accuracy.

45. Discuss document structure analysis in large datasets.

Document structure analysis identifies headings, sections, paragraphs, and rhetorical relations in text. Large datasets pose challenges like inconsistent formatting, noisy text, multiple languages, and domain-specific conventions. Efficient algorithms must scale, handle diverse document types, and extract hierarchical structures accurately. Rule-based methods may fail on heterogeneous data, while machine learning models require feature extraction and training on annotated corpora. Neural models, especially transformers, can process large texts and learn structural patterns automatically. Accurate document structure analysis is crucial for summarization, information retrieval, knowledge extraction, and AI document understanding.

46. Explain evaluation metrics for parsing accuracy.

Parsing accuracy is evaluated using metrics like **precision**, **recall**, and **F1-score**, measuring how correctly predicted structures match gold-standard annotations. **Attachment score** is used in dependency parsing to check if words are linked to correct heads. **Labeled vs. unlabeled**

metrics distinguish between correctly predicted relations and merely structural correctness. For probabilistic parsers, **perplexity** or **log-likelihood** measures model confidence. Evaluating on multiple datasets ensures robustness and generalization. Challenges include ambiguous sentences, multiple valid parses, and cross-lingual consistency. Accurate evaluation guides parser improvement and deployment in real-world NLP tasks.

47. Discuss statistical approaches to word sense disambiguation.

Statistical WSD uses probabilities and context to predict word meanings. Methods include **Naive Bayes**, **decision trees**, and **maximum entropy models**, which learn from labeled corpora. Features may include surrounding words, POS tags, syntactic patterns, or collocations. For example, in “The bank raised interest rates,” context words like “interest” and “rates” indicate the financial sense. Statistical WSD handles large datasets and unseen combinations better than rule-based systems. Challenges include sparse data, domain variation, and rare senses. Combining statistical WSD with knowledge-based or neural methods improves accuracy.

48. Explain language model overfitting and regularization.

Overfitting occurs when a language model learns the training data too well, capturing noise and rare patterns, which reduces performance on new texts. Signs include high training accuracy but poor validation performance. Regularization techniques prevent overfitting, such as **dropout**, **weight decay**, and **early stopping**. Data augmentation and training on diverse corpora also help. Neural LMs, especially large transformers, are prone to overfitting on small or domain-specific datasets. Proper regularization ensures generalization, reliable predictions, and better performance in NLP applications like text generation and machine translation.

49. Discuss the future of cross-lingual NLP.

Cross-lingual NLP focuses on building systems that understand and process multiple languages. Future developments include **universal multilingual models** with better zero-shot learning for low-resource languages, **improved transfer learning** across language families, and **context-aware cross-lingual embeddings**. Challenges like varying scripts, idioms, and cultural nuances remain. Integration with large pre-trained transformer models will enhance machine translation, cross-lingual search, and multilingual AI assistants. Research on aligning semantic and syntactic representations across languages will further advance the field. Cross-lingual NLP aims to bridge language barriers and enable global AI applications.

50. Explain how discourse processing improves text understanding in AI systems.

Discourse processing analyzes relationships between sentences, paragraphs, and the overall text. It includes reference resolution, coherence and cohesion analysis, discourse markers, and discourse structure. This allows AI systems to track entities, maintain context, understand logical relations, and infer implicit information. For example, resolving “he” or “it” across sentences ensures accurate comprehension. Discourse-aware systems improve machine translation, summarization, dialogue systems, and question answering. Challenges include long-distance dependencies, topic shifts, and informal text structures. Effective discourse processing enables AI to understand text like humans, enhancing reasoning and context-aware responses.

